EXPERIMENT:14

AIM:

To write a C program to implement the one-time pad version of the Vigenère cipher for encryption and decryption.

CODE:

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>


int main() {

   char p[] = "send more money";

   int k[] = {9,0,1,7,23,15,21,14,11,11,2,8,9};

   int i, j = 0, c[50];


   printf("Ciphertext: ");

   for(i = 0; p[i]; i++) {

      if(isalpha(p[i])) {

         c[j] = (tolower(p[i]) - 'a' + k[j]) % 26;

         printf("%c", 'a' + c[j]);

         j++;

      } else printf(" ");

   }


   printf("\nKey for 'cash not needed': ");

   char q[] = "cash not needed";

   j = 0;

   for(i = 0; q[i]; i++) {
```

```
        if(isalpha(q[i])) {
            printf("%d ", (c[j] - (tolower(q[i]) - 'a') + 26) % 26);
            j++;
        }
    }
    return 0;
}
```

OUTPUT:

